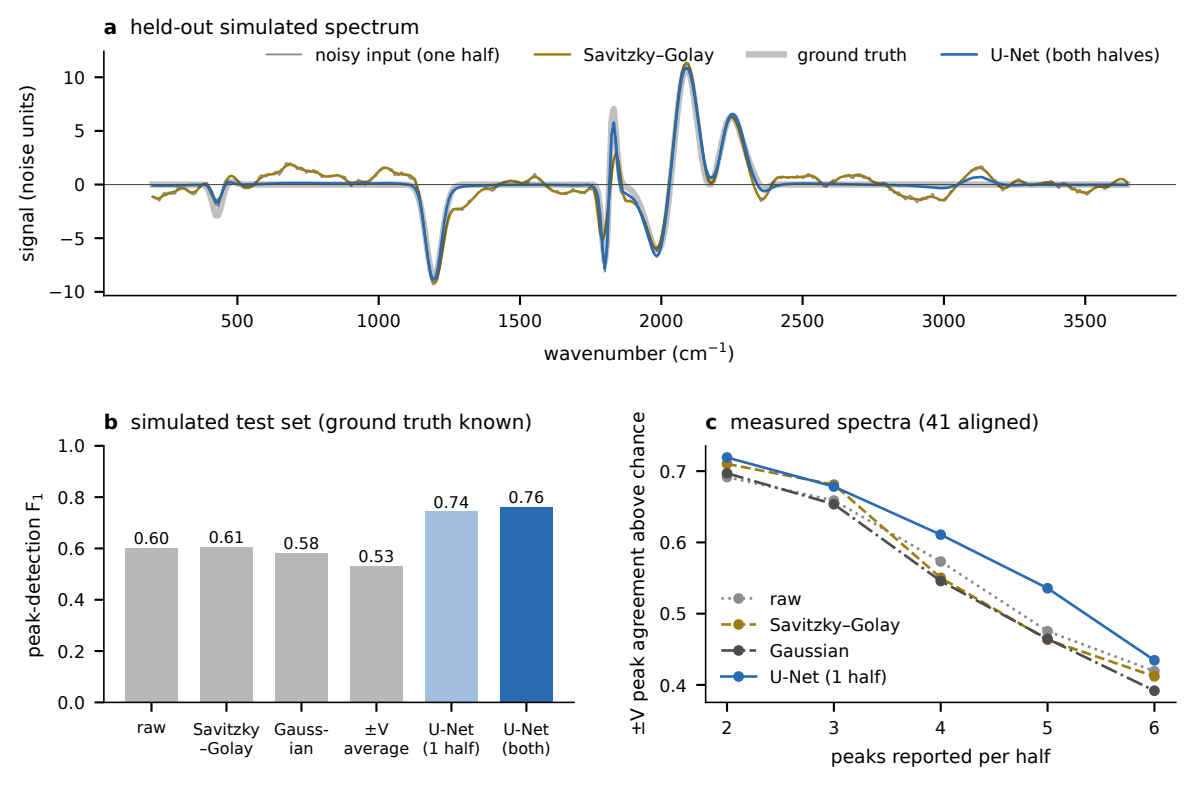
a held-out simulated spectrum
noisy input (one half)
Savitzky–Golay
ground truth
U-Net (both halves)
signal (noise units)
wavenumber (cm^{-1})
b simulated test set (ground truth known)
peak-detection F_1
0.60
0.61
0.58
0.53
0.74
0.76
raw
Savitzky –Golay
Gauss- ian
±V average
U-Net (1 half)
U-Net (both)
c measured spectra (41 aligned)
±V peak agreement above chance
peaks reported per half
raw
Savitzky–Golay
Gaussian
U-Net (1 half)